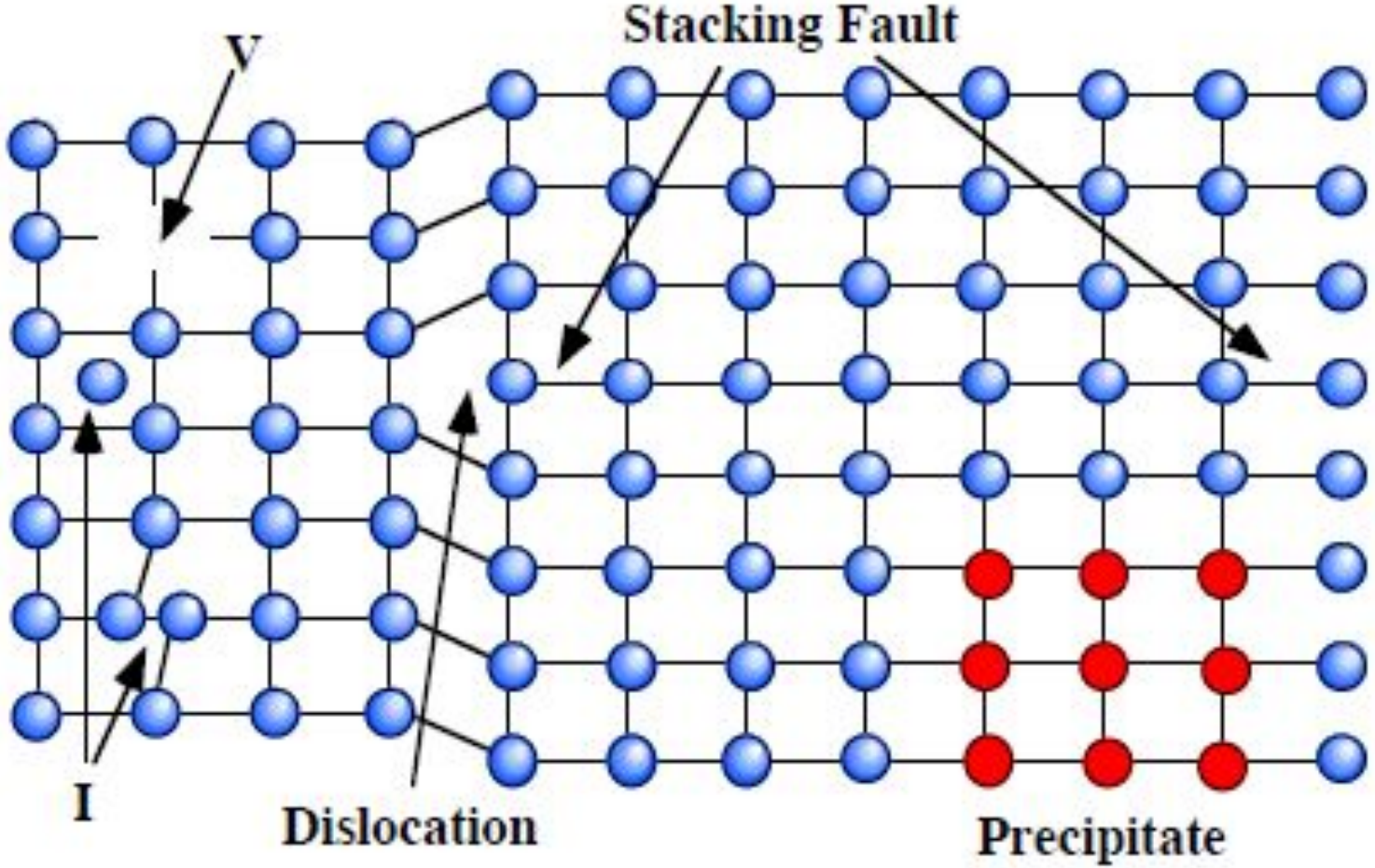
EE2021 -
Materials Science for Electrical Engineers

Defects in Crystal

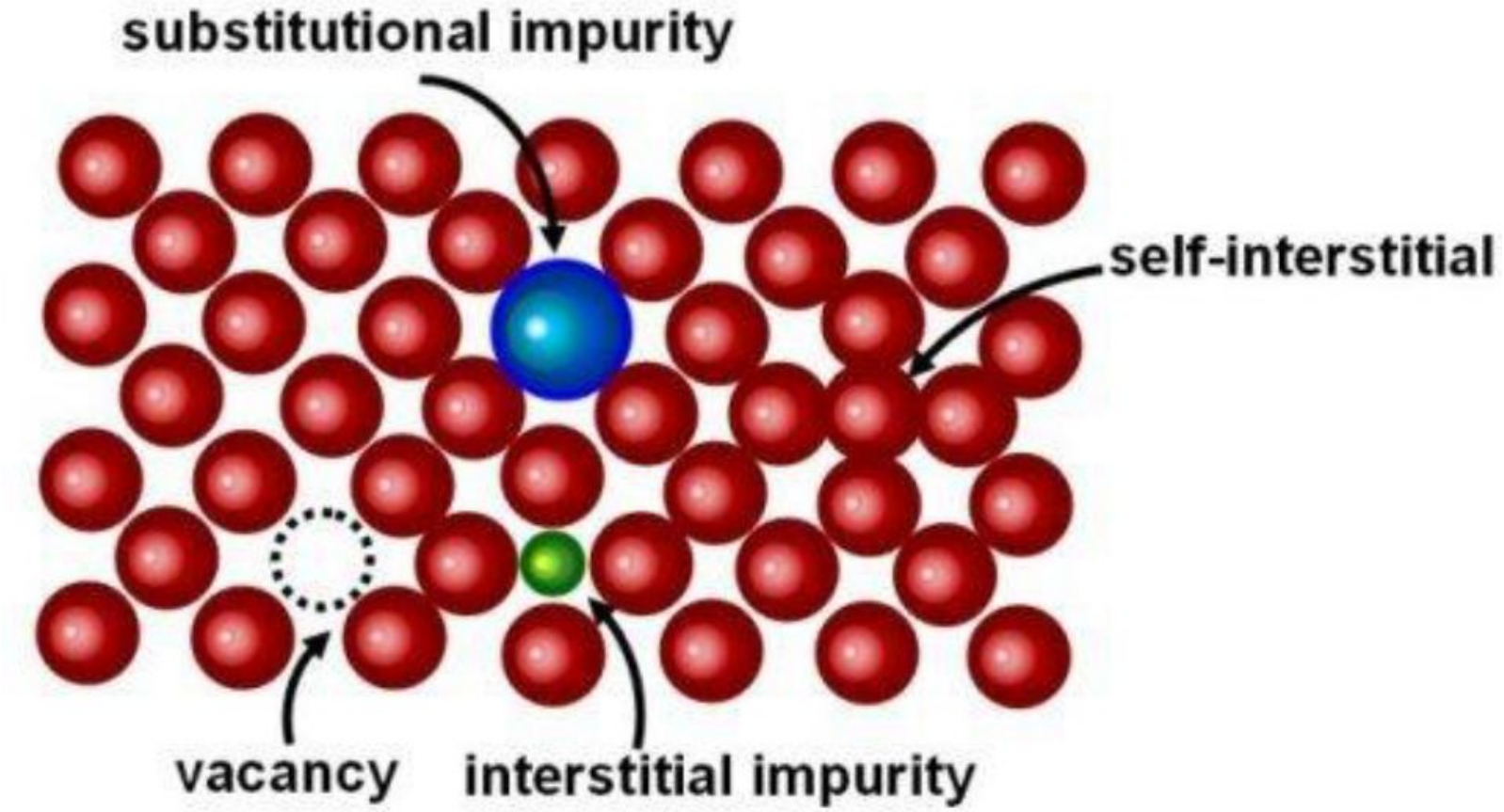
Deleep R. Nair & Sayak Dutta Gupta
Department of Electrical Engg, IIT Madras

Imperfections in Crystal

- Crystal Structure: perfect order in crystalline materials on an atomic scale
- Such an idealized solid does not exist → contain large numbers of various defects or imperfections
- Crystalline defect refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter
  1. **Point defects** (vacancies, interstitial defects, substitution defects)
  2. **Line defects** (screw dislocation, edge dislocation)
  3. **Area defects** (Stacking fault (insertion or removal of plane), material surface, grain boundaries)
  4. **Volume defects -** Voids, precipitates
- Defects could be formed during growth/deposition, subsequent process to create structures or device operation
- Influence of defect on material properties is not always adverse → specific characteristics are deliberately fashioned by introduction of controlled amounts of particular defects


Stacking Fault
V
I
Dislocation
Precipitate

0D: Point Defects



Point Defects - Vacancies & Self-Interstitials

Self-interstitial –

- A self-interstitial is an atom from the crystal that is crowded into an interstitial site, a small void space that under ordinary circumstances is not occupied.
- Formation of this defect is not highly probable → exists in very small conc.

Vacancy -

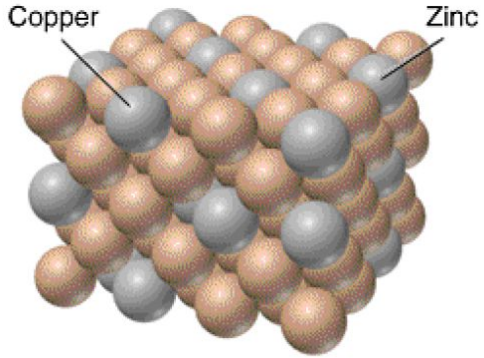
- A vacancy is produced when an atom is missing from a normal site
- The equilibrium number of vacancies N_v or a given quantity of material

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

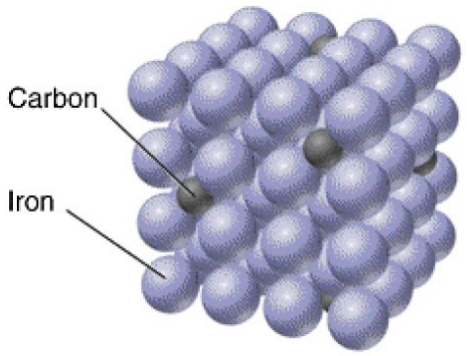
N is the total number of atomic sites (per m^3),
Q_v is the energy required for the formation of a vacancy (J/mol or eV/atom)
T is absolute temperature
k is Boltzmann's constant

Point Defects - Impurities in Solids

- A pure metal consisting of only one type of atom just isn't possible
- Impurity or foreign atoms will always be present (near/below the detection limit), and some will exist as crystalline point defects
- The addition of impurity atoms to a metal will result in the formation of a solid solution
- Impurity point defects found in solid solutions are of two types:
  1. **Substitutional**–one atom is replaced by a different type of atom
  2. **Interstitial**–extra atom is inserted into the lattice structure at a normally unoccupied position



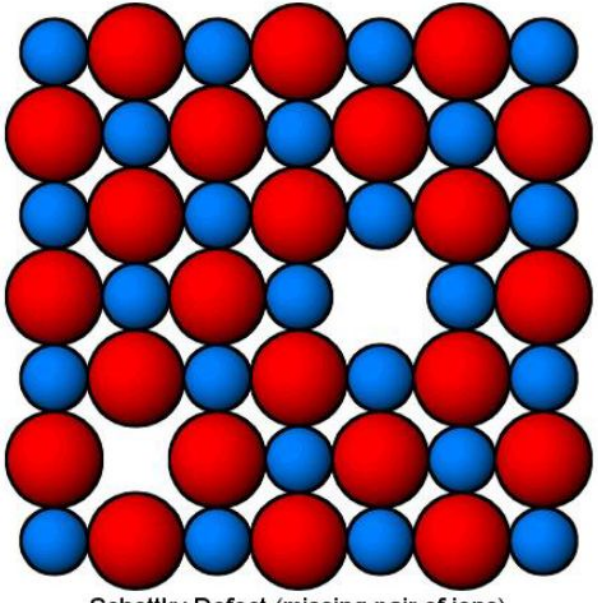
A Brass, a substitutional alloy



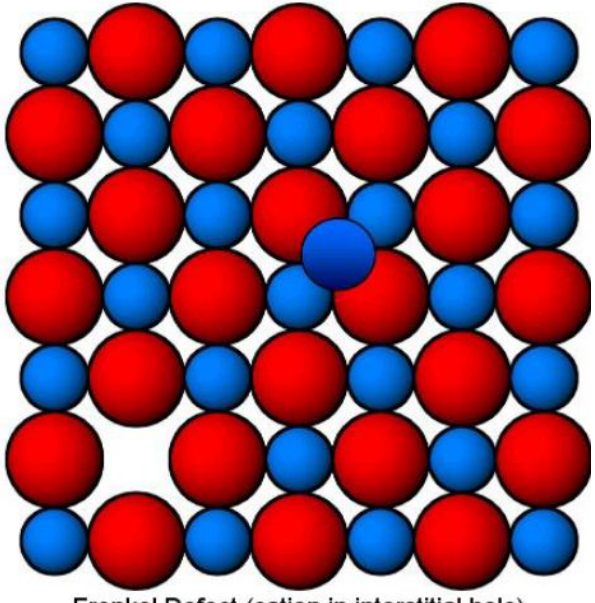
B Carbon steel, an interstitial alloy

Other Point Defects

- Frenkel defect – ion jumps from a normal lattice point to an interstitial site, leaving behind a vacancy (vacancy-interstitial)
- Schottky defect – pair of vacancies in ionically bonded material. Both anion and cation must be missing from the lattice to maintain the crystal neutral

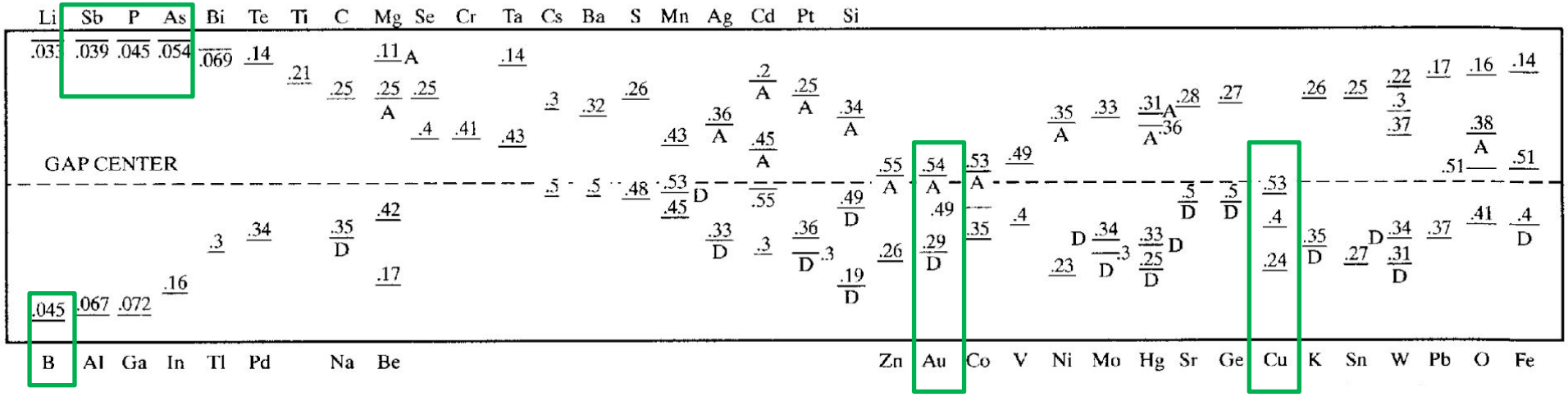


Schottky Defect (missing pair of ions)



Frenkel Defect (cation in interstitial hole)

Point Defects: Dopants vs Traps



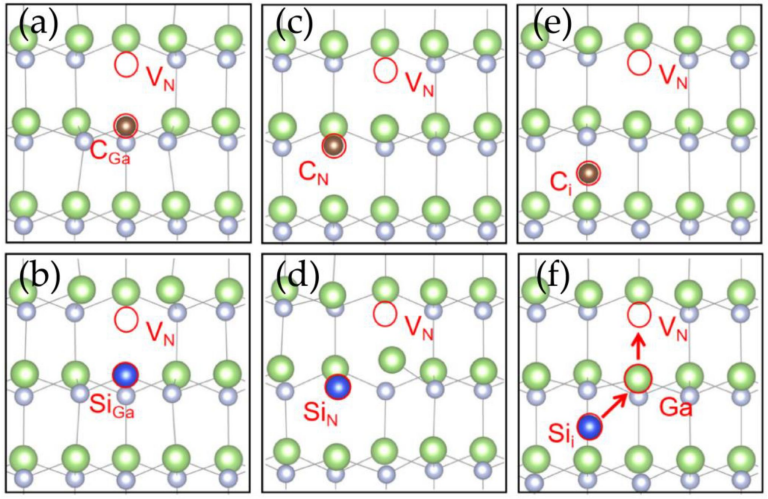
Donor (Acceptor) energy levels should be close to CB (VB) edge

Traps are localized (in space also) energy levels near the midgap. They can capture/emit electron or hole. Determines switching speed and leakage current

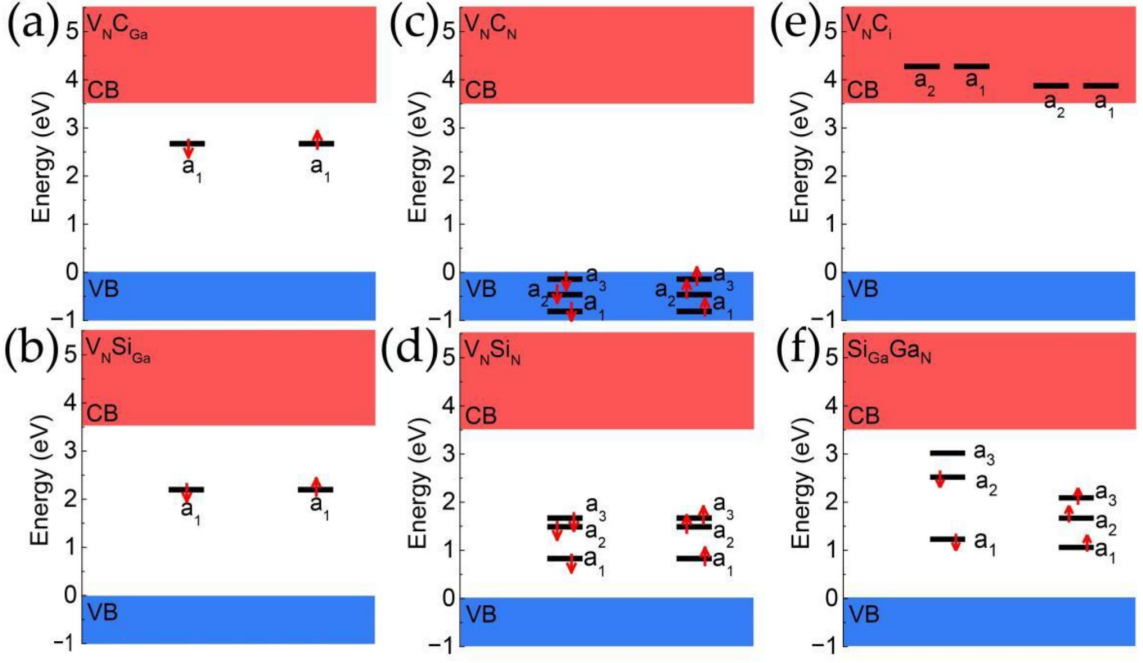
CDC 6600 (1st supercomputer) was made with 600k gold diffused transistors

https://www.computerhistory.org/siliconengine/silicon-transistor-exceeds-germanium-speed/

Benefits of Point Defects: Doping and Trap states



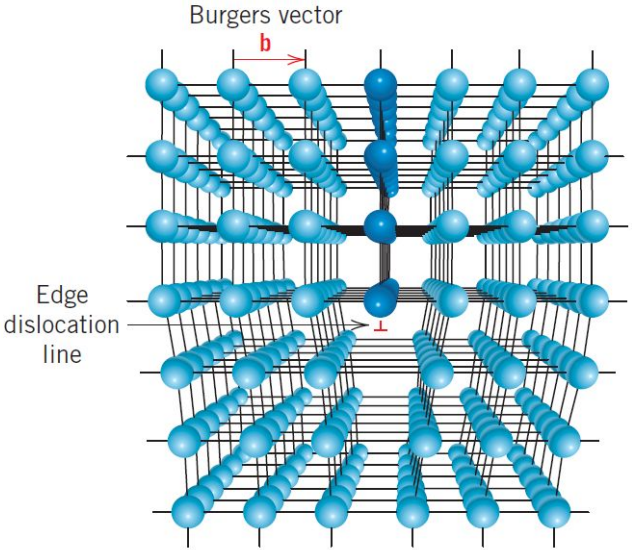
Carbon and Silicon Impurity Defects in GaN

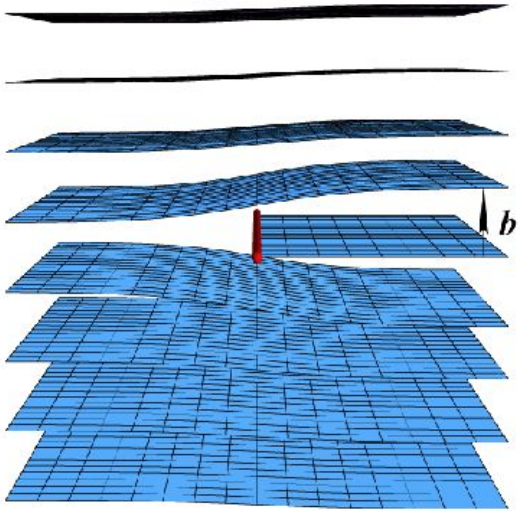


(1D) Line Defects: Dislocations

Edge dislocation:

- Extra portion of plane of atom / half-plane, edge of which terminates within the crystal
- Centers on the dislocation line (along the end of extra half-plane of atoms

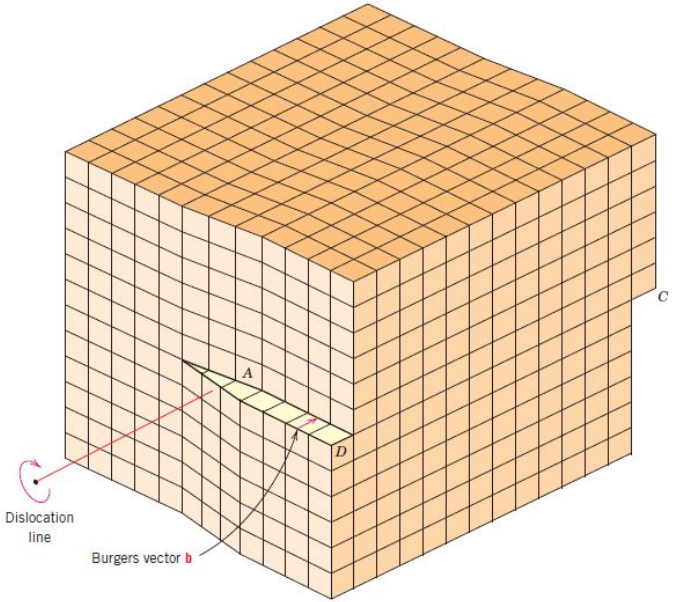


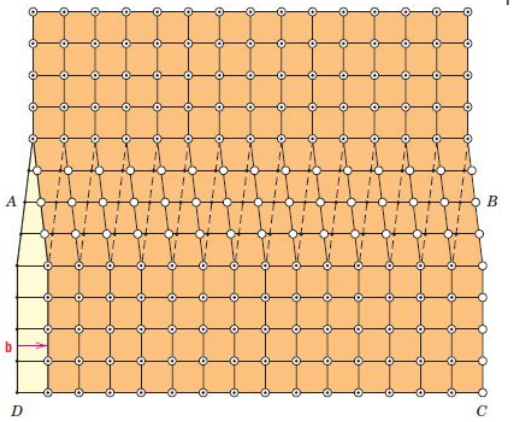


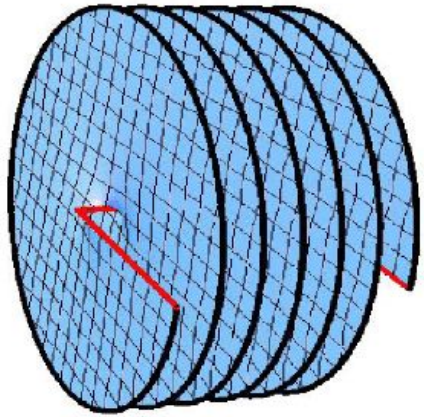
(1D) Line Defects: Dislocations

Screw dislocation:

- Path spirals around dislocation line penetrating through otherwise parallel planes
- Upper front region of crystal shifted one atomic distance to the right relative to bottom portion

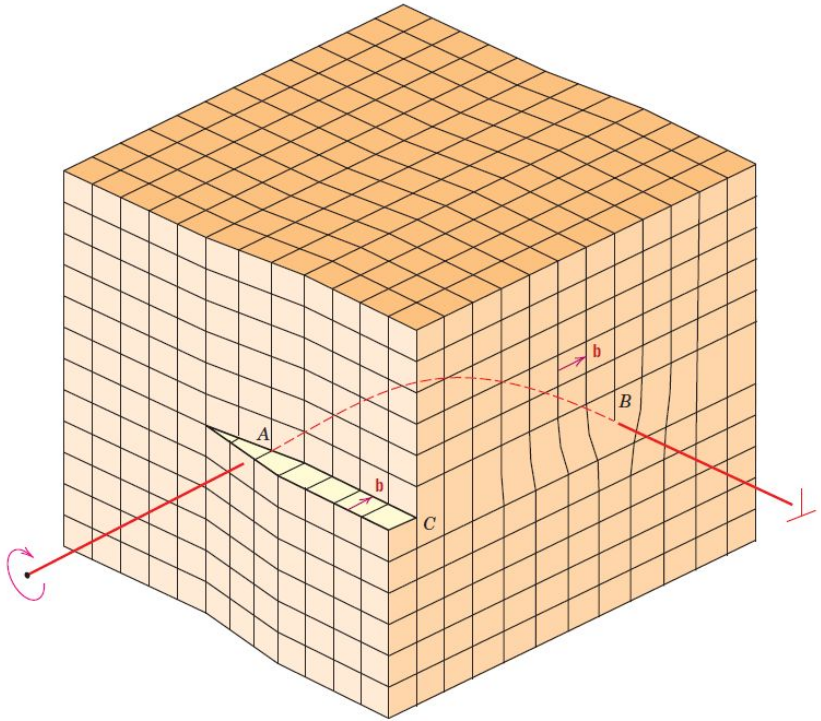


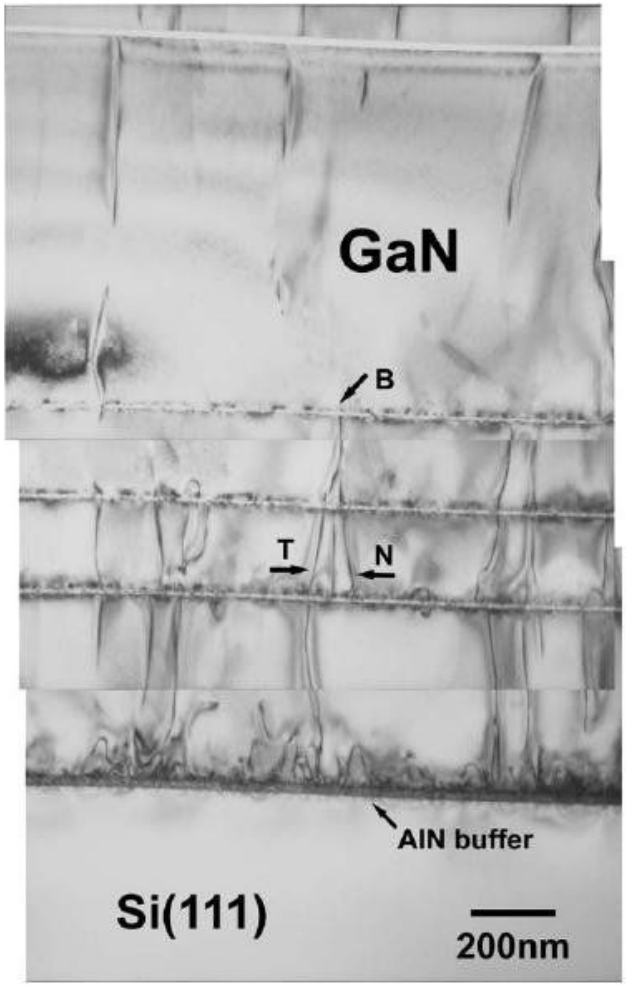




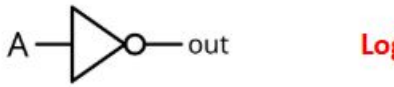
Mixed Dislocations

- Most dislocations found in crystals neither pure edge nor pure screw
- Exhibit components of both types → mixed dislocations

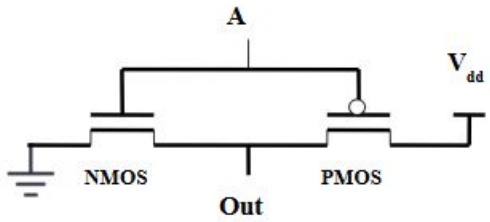




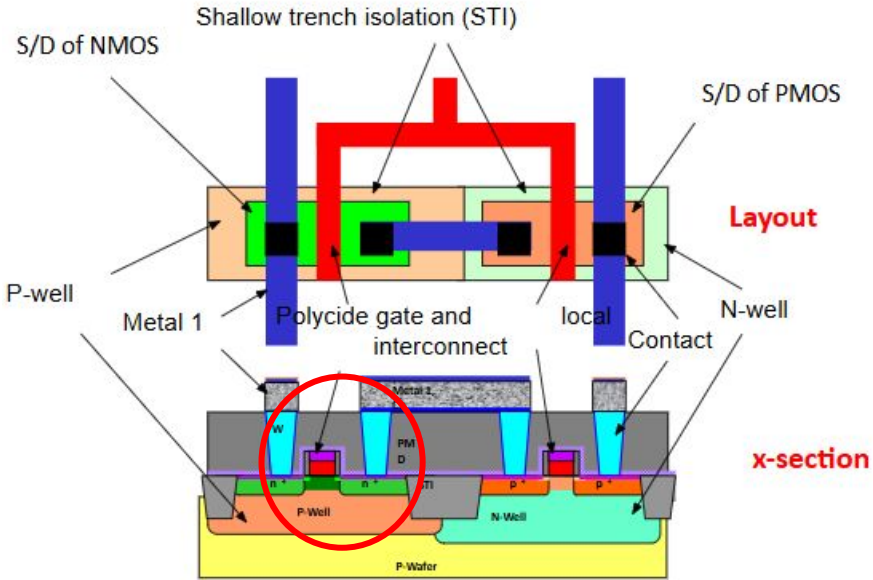
A friendly stacking fault



Logic gate

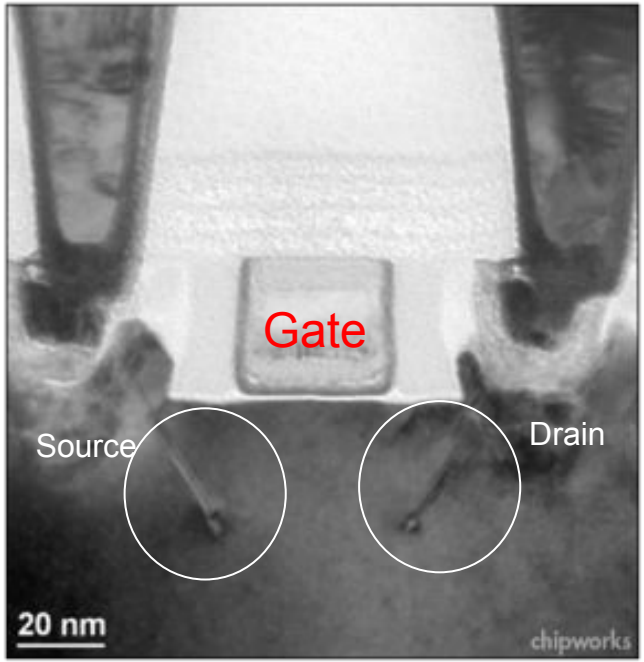


Circuit Schematic



Layout

x-section



Intel 32nm Westmere µP (2010)

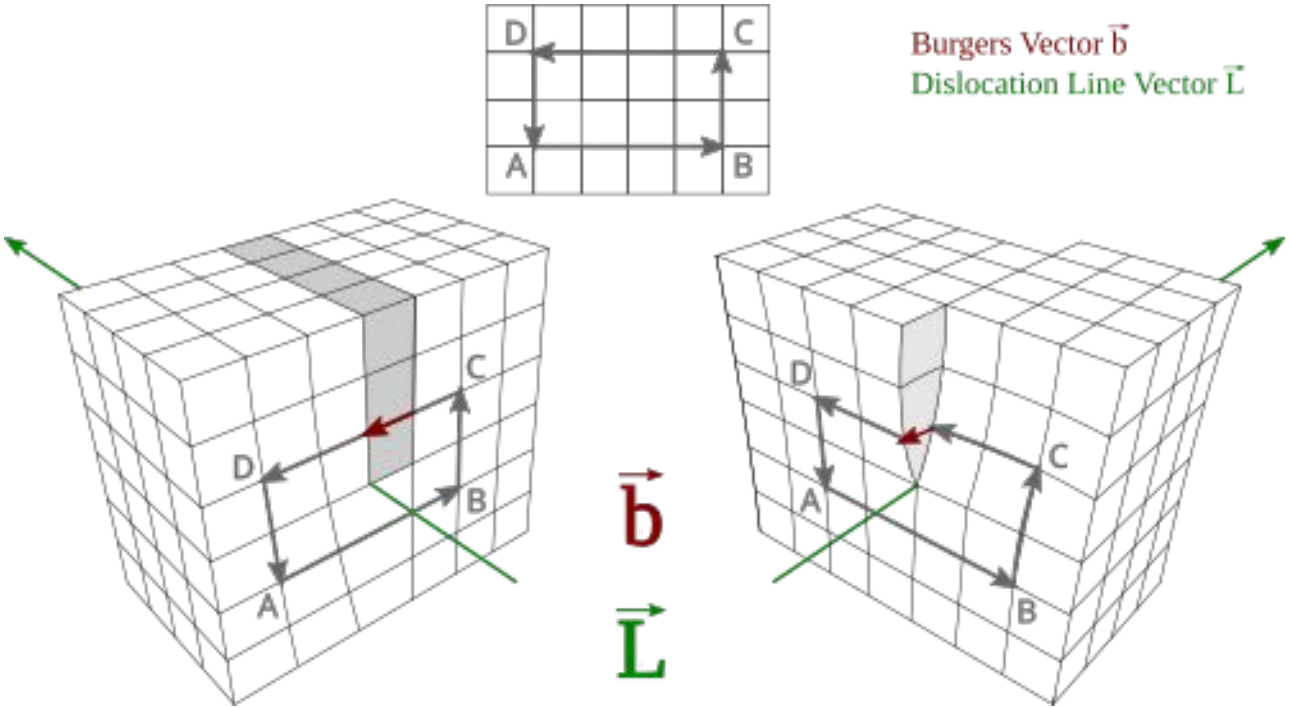
Carefully created dislocation to improve the mobility of electrons and drive current of transistor

https://www.semiconductor-digest.com/intel-clarifies-32nm-nmos-stress-mechanism-at-iedm-2011/

Burgers vector

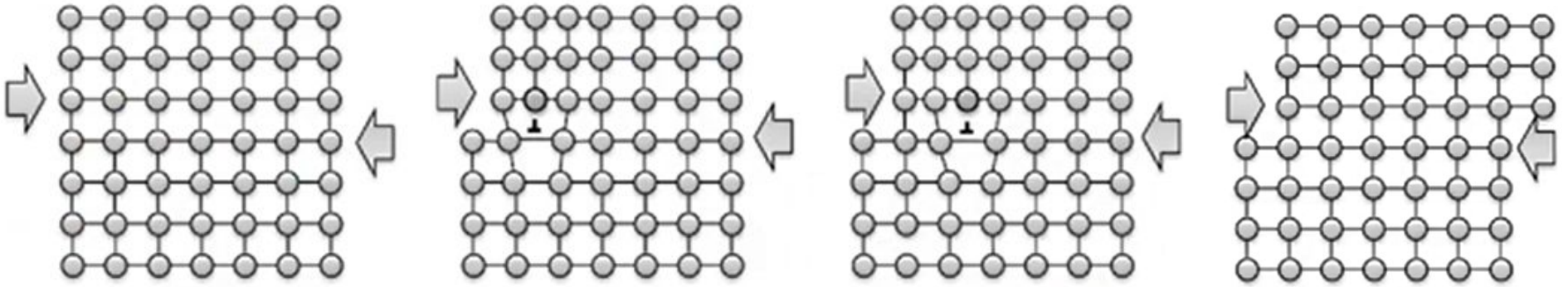
- Burgers vector : Displacement vector that “closes the loop” when traversing an equal number of lattice steps around the defect

- Edge dislocation: Burgers vector is perpendicular to dislocation line
- Screw dislocation: Burgers vector is parallel to the dislocation line

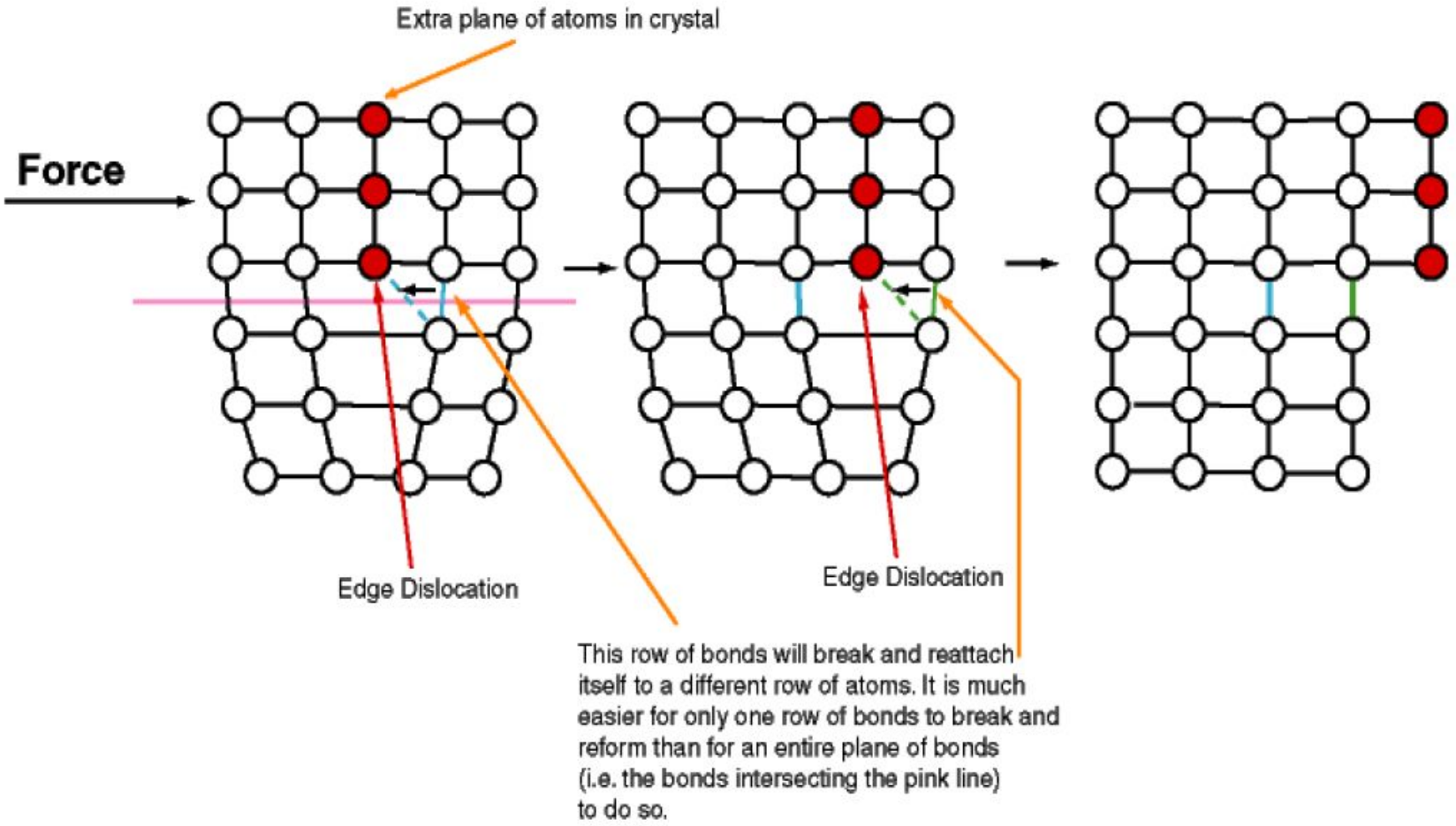


Atom/Lattice movement

- For mobile dislocation, the direction of the vector is usually on one of the most packed crystallographic planes (slip planes) and always in the most packed crystallographic directions (slip directions).
- Slip plane is plane containing both Burgers vector & dislocation line
- Slip planes + slip directions are called **slip systems** (represent the planes and the directions in which plastic deformation can occur)



Dislocation movement



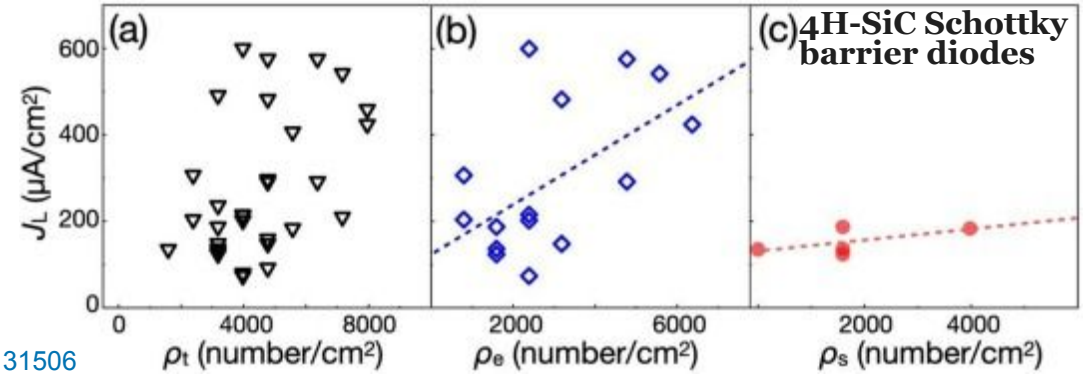
Effects of dislocation

- **Advantage: Facilitate plastic deformation**:
  a. Step movement of atomic planes instead of all atomic bonds breaking
  b. Deformation possible at much lower stresses → Enable material to bend and stretch without breaking
- **Advantage: Increase strength and hardness**:
  a. Movement of dislocations causes material to deform
  b. Strength and hardness of material can be tuned by Dislocation density
  c. Higher dislocation density → harder for dislocations to move → leading to increased strength and hardness (work hardening)
- **Disadvantage:** Increased leakage in electronic devices

ρ_t: total dislocation density

ρ_e: edge dislocation density

ρ_s: screw dislocation density



https://doi.org/10.1016/j.matlet.2021.131506

2-D Planar Defects (Grain Boundaries)

- Boundaries that have two dimensions and separate regions of materials with different crystal structures and/or orientations
- Imperfections include external surfaces, grain boundaries, phase boundaries, twin boundaries, and stacking faults

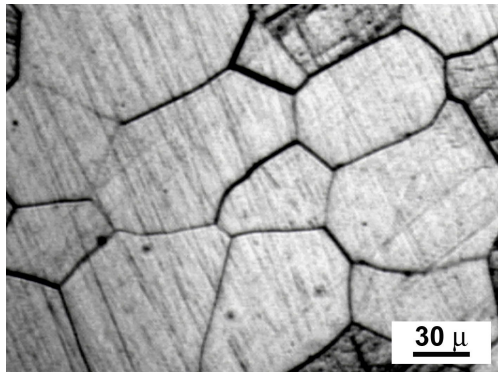
External Surfaces:

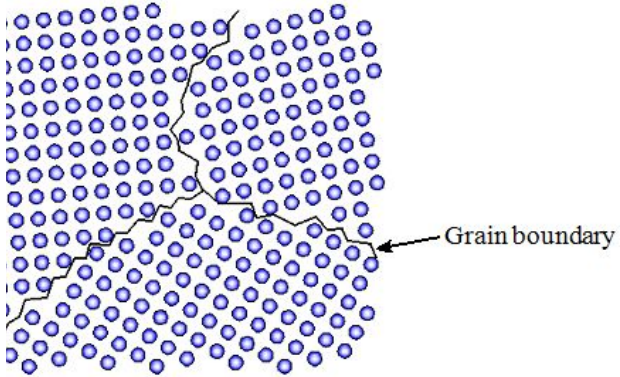
❑ Boundaries along which the crystal structure terminates

❑ Surface atoms not bonded to the maximum number of nearest neighbors → in a higher energy state than the atoms at interior positions

Grain Boundaries:

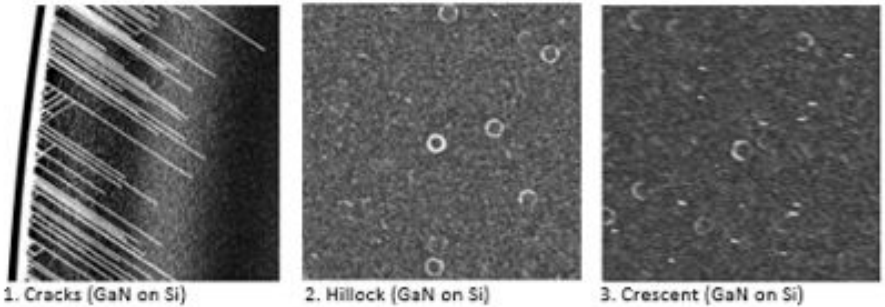
2-D interface between adjacent grains (single crystals) in a polycrystalline material

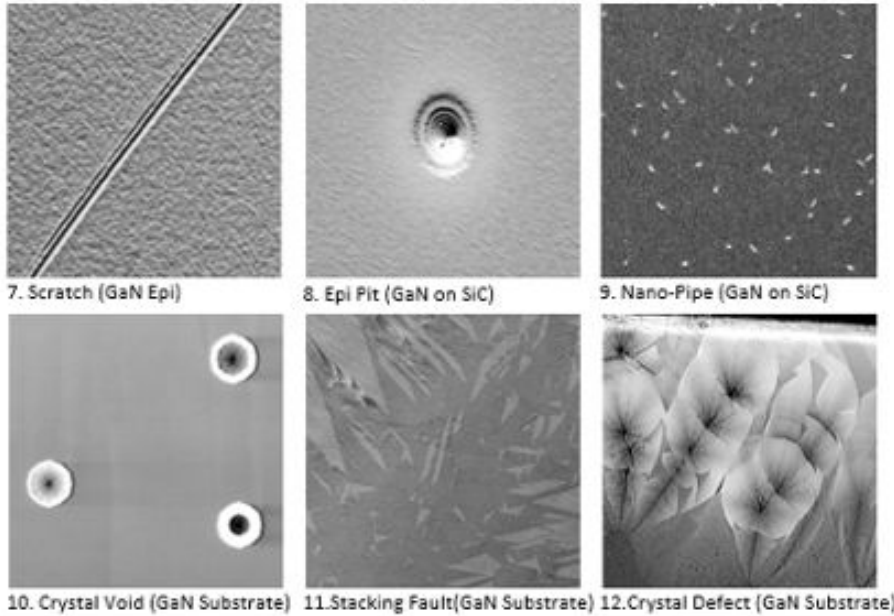




3-D Bulk or Volume Defects

- Include pores, cracks, foreign inclusions, and other phases
- Normally introduced during processing and fabrication steps


1. Cracks (GaN on Si)
2. Hillock (GaN on Si)
3. Crescent (GaN on Si)
4. Showerhead (GaN on PSS)
5. Hex Bump (GaN on Sapphire)
6. Epi Bump (GaN on Sapphire)


7. Scratch (GaN Epi)
8. Epi Pit (GaN on SiC)
9. Nano-Pipe (GaN on SiC)
10. Crystal Void (GaN Substrate)
11.Stacking Fault(GaN Substrate)
12.Crystal Defect (GaN Substrate)
13. PL Dark (GaN Substrate)
14. Hex Pit (GaN on GaN)
15.Triangular Hillock (GaN on GaN)